Area and surface area

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. Find 3^2. Write the repeated factors.

6. Write a story for $\frac{5}{6} \div \frac{1}{3}$. Solve it.

7. $48 \div 4$

8. Find the surface area of a closed 3 by 3 by 4 cm box.

9. Find the surface area of a closed 2 by 3 by 5 cm box.

10. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

11. A square pyramid has base side 4 cm and triangular-face height 3 cm. Find surface area.

12. A closed cube has side 2 cm. Jo counts five faces and gets 20 square cm. Repair the total.

13. Ava finds a cube's surface area by adding five faces. Each face is 9 square cm. She gets 45. Explain and repair it.

14. A rectangle has length 9 cm and width 6 cm. Find area and perimeter.

15. A parallelogram has base 7 cm, height 6 cm, and slanted side 8 cm. Find area and name the unused length.

16. A right triangle has perpendicular legs 14 cm and 6 cm. Find area using either leg as base.

17. A triangle has area 42 square cm and base 14 cm. Find its matching height.

18. A10 cm by 8 cm rectangle loses a3 cm by 2 cm corner. Find remaining area.

CHALLENGE / Explain your plan and check the result

C1. Two closed boxes have dimensions 2 × 3 × 6 cm and 3 × 3 × 4 cm. Find each volume and surface area. Which uses less cardboard for the same capacity, and by how much?

C2. A square pyramid has base side 8 cm and triangular face height 5 cm. Only its four triangular faces are painted. Paint costs $0.15 per square cm. Find the paint cost; then find the extra cost if the base is painted too.